

Sequences and series



1 A sequence is a list of terms, and a series is the sum of the terms.

2 6

3 196

4 a_5

5

a Finite

b Infinite

c Finite

d Infinite

6 Finite

7

a Infinite

b Finite

8

a 0, 3, 6, 9, 12

b $-2, -4, -8, -16, -32$

c 4, 8, 16, 32, 64

d $-6, 24, -72, 192, -480$

e 1, 0.7071, 0.5774, 0.5, 0.4472

f 2, 2.25, 2.3704, 2.4414, 2.4883

9

a $\frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}$

b $\frac{2}{3}, 1, 1, \frac{19}{21}$

c $-1, -\frac{1}{4}, 0, \frac{1}{16}$

d 0, $-1, 2, -3$

10

a 4, 3, 4, 11

b 3, 5, 8, 12

c $-2, -2, 0, 8$

d 3, 5, 15, 75

e 3, 2, 0, -8



Sigma notation

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a

$$\sum_{k=1}^{\infty} 4k$$

b

$$\sum_{k=1}^6 (-3)^k$$

c

$$\sum_{k=1}^5 \frac{1}{k^3}$$

d

$$\sum_{k=1}^7 \frac{1}{3k}$$

e

$$\sum_{k=1}^{11} \frac{2}{5+k}$$

f

$$\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$$

12

a 40

b $\frac{47}{60}$

c $\frac{19}{20}$

d 15

e 14

f 1815

g 368

h -16

i $\frac{63}{2}$

j 38

13 20

14 -10

15 30

16 -8

17

a -6

b -8

18

a -54

b 65

19



a

$$\sum_{i=1}^7 (i + i^3) = 2 + 10 + 30 + 68 + 130 + 222 + 350$$

$$\begin{aligned} \sum_{i=1}^7 i + \sum_{i=1}^7 i^3 &= (1 + 2 + 3 + 4 + 5 + 6 + 7) + (1 + 8 + 27 + 64 + 125 + 216 + 343) \\ &= 2 + 10 + 30 + 68 + 130 + 222 + 350 \end{aligned}$$

b Yes

c Yes

Applications

20 $m_1 = 2000, m_n = (0.76) m_{n-1}$

21

a \$1, \$3, \$7, \$15

b \$38, \$43, \$48, \$53

c Day 6